

Docker overview

1. Docker Concepts

Topic	Description
Docker Image	A packaged blueprint containing an application and its dependencies.
Docker Container	A running instance of a Docker image.
Dockerfile	A file containing instructions to build a Docker image.
Docker Volume	Stores data persistently, even when a container is removed.
Bind Mount	Connects a host computer's folder or file to a container.
Docker Compose	Defines and manages multiple containers using a YAML file.
Port Mapping	Exposes a container's port through the host machine.
Docker Hub	An online registry for storing and sharing Docker images.
WSL Integration	Allows Docker to work with Linux commands and environments on Windows.

Docker Commands

1. Docker Image Management

- `docker --version` → Check Docker version.
- `sudo docker images` → List available images.
- `sudo docker build -t authentication-api:1.0 .` → Build an image
- `sudo docker rmi IMAGE_NAME` → Remove an image
- `sudo docker pull IMAGE_NAME` → Download an image from Docker Hub.
- `sudo docker push IMAGE_NAME` → Upload an image to Docker Hub.

2. Docker Container Management

- `sudo docker run -d --name auth-api -p 8000:8000 --env-file .env authentication-api:1.0` → Create and run a container in the background with port mapping and environment variables.
- `sudo docker ps` → List running containers.
- `sudo docker ps -a` → List all containers, including stopped ones.
- `sudo docker start auth-api` → Start a stopped container.
- `sudo docker stop auth-api` → Stop a running container.

- Stop a running container. → Restart a container.
- `sudo docker rm auth-api` → Remove a stopped container.
- `sudo docker rm -f auth-api` → View container logs.
- `sudo docker logs -f auth-api` → Follow container logs in real time.
- `sudo docker exec -it auth-api /bin/sh` → Open a shell inside a running container.
- `sudo docker inspect auth-api` → View detailed container information.
- `sudo docker stats` → Monitor container CPU, memory, and resource usage.

3. Docker Volume Management

- `sudo docker volume ls` → List Docker volumes.
- `sudo docker volume create my-volume` → Create a named volume.
- `sudo docker volume inspect my-volume` → View volume details.
- `sudo docker volume rm my-volume` → Remove a volume.
- `sudo docker volume prune` → Remove unused volumes.

4. Docker Compose Commands

- `sudo docker compose up -d` → Create and start services in the background.
- `sudo docker compose down` → Stop and remove Compose containers and networks.
- `sudo docker compose ps` → List Compose services and their status.
- `sudo docker compose logs` → View service logs.
- `sudo docker compose build` → Build or rebuild service images.
- `sudo docker compose restart` → Restart Compose services.
- `sudo docker compose down -v` → Stop and remove containers, networks, and associated Compose-managed volumes. Warning: This can delete persistent data.

5. Dockerfile

- You created a Dockerfile to package your application.

```
</> dockerfile

FROM python:3.12-slim

WORKDIR /app

COPY requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 8000

CMD ["python", "server.py"]
```

- FROM ---> Selects the base image.
- WORKDIR ---> Sets the working directory.

- COPY ---> Copies files into the image.
- RUN ---> Executes commands during image building.
- EXPOSE --> Documents the application's listening port.
- CMD ---> Defines the default startup command.

6. Docker Ignore

```
.env
.git
__pycache__
*.pyc
```

- It prevents unnecessary files and sensitive configuration from being included in the Docker build context.

7. Port Mapping

- `sudo docker run -d --name auth-api -p 8000:8000 authentication-api:1.0`
- - 8000 (first) → Host machine port.
 - 8000 (second) → Container port.
- Allows requests to reach the API running inside the container.